Wingman: A Volatility-Surface Trading Agent for SPY Options

Paper account: `PA3ICB1OURR1` — github.com/[repo]

Overview

Wingman is an autonomous SPY options-trading pipeline built on Alpaca's paper-trading API. Each cycle it fetches a live SPY option-chain snapshot, fits a two-component lognormal mixture model (Brigo–Mercurio) to the market-implied volatility smile, compares the model's price at each strike against the market's, and proposes a defined-risk structure – a long straddle where the market is cheap versus the model, or a short call vertical where it is rich – subject to a chain of materiality and liquidity gates before any order is built.

AI / Decision Logic

The core signal is a residual, not a raw IV level: implied vols are recovered from live bid/ask mids via Black–Scholes inversion, fit in price space to the mixture model via nonlinear least squares (Levenberg–Marquardt), and the fitted model's price at each strike is compared against the market's call-equivalent mid. A residual that clears every gate below becomes the highest-conviction trade candidate for that cycle.

A planned second layer asks an LLM (Gemini) to classify the current market regime and gate proposed trade sizing (full / half / stand-down) before execution.

Risk Gates

Every proposal must clear all of the following before an order is built:

- **Degenerate-fit guard**: both fitted volatilities must sit inside an economically plausible SPY vol band; a fit against its numerical bounds is rejected outright rather than trusted as signal.
- **Liquidity filter**: strikes are only considered if both legs clear a minimum open-interest threshold and carry a genuine two-sided quote.
- **Tradable moneyness band**: candidates are restricted to strikes close to spot, since the unshifted symmetric mixture cannot reproduce SPY's actual skew and would otherwise misprice the wings by construction, not by real mispricing.
- **Materiality gates**: a residual must exceed both a multiple of the strike's own quote half-spread and a multiple of the fit's own RMSE noise floor – otherwise it is indistinguishable from quote noise or estimation error.
- **Execution-spread gate**: applied again on the *actual* legs to be traded (which can be materially wider than the reference quote used for the residual), so a wide real spread cannot eat a signal that looked clean on paper.
- **Structural risk control**: every short leg is submitted only inside a multi-leg (mleg) order alongside its covering long leg, so a naked short position can never exist even transiently.
- **Dry-run default**: order submission defaults to a safety dry run; live submission requires an explicit, non-default environment flag.

Alpaca Infrastructure

Market data (spot trades, option snapshots with bid/ask/IV/greeks) and reference data (option contracts, open interest, the market clock) are pulled via the `alpaca-py` SDK against Alpaca's paper endpoints. Order submission runs through the Alpaca CLI: a multi-leg order payload is written to a temp file and piped into `alpaca api POST /v2/orders`, the confirmed-reliable path for mleg orders on this platform. Account equity and every open position are snapshotted once per cycle for a mark-to-market equity time series alongside the decision log.

Wingman is an in-progress build: the mixture-fit and gating pipeline runs live and has produced real (paper) decisions this week, while the LLM-based regime gate is still under active development.